Problem-Solution Fit

Solution Architecture

Date	11 July 2026
Team ID	
Project Name	Personalized Networking Assistant
Maximum Marks	5 Marks

PROJECT OVERVIEW
Networking Assistant
Project Description

Networking Assistant is an AI-powered web application designed to help students, professionals, and job seekers prepare for networking events more effectively. The system uses Generative AI to analyze event details, suggest relevant conversation topics, verify factual information, and maintain a history of previous interactions.

The application provides users with intelligent recommendations before attending conferences, workshops, seminars, career fairs, and professional meetups. By leveraging Artificial Intelligence, the platform improves communication, increases confidence, and enables users to build stronger professional relationships.

The project is built using a FastAPI backend, a Streamlit frontend, and Google's Gemini Generative AI model for intelligent content generation.

PROBLEM STATEMENT

Attending networking events can be challenging, especially for students and professionals who struggle to initiate meaningful conversations or identify relevant discussion topics. Many participants lack confidence, have difficulty verifying information shared during conversations, and fail to maintain records of important networking interactions.

Traditional networking preparation methods are manual, time-consuming, and often ineffective. There is a need for an AI-powered solution that assists users in preparing for networking events by generating intelligent conversation topics, verifying factual information, and organizing networking history.

PROPOSED SOLUTION

The proposed solution is an AI-powered Networking Assistant that simplifies networking preparation through intelligent automation.

The system accepts an event description from the user and analyzes it using Google's Gemini AI model. Based on the event context, the application generates personalized networking topics, suggests conversation starters, performs fact verification, collects user feedback, and stores interaction history for future reference.

The solution provides a simple and interactive web interface where users can prepare for networking events quickly and efficiently.

OBJECTIVES

The primary objectives of the project are:

- Develop an AI-powered networking assistant.
- Analyze networking event descriptions.
- Generate intelligent conversation topics.
- Verify factual information using AI.
- Maintain networking history.
- Collect user feedback for continuous improvement.
- Provide an easy-to-use web application.
- Improve users' networking confidence and communication skills.

FEATURES

- AI-Based Event Analysis
- Intelligent Topic Generation
- Conversation Starter Suggestions
- AI Fact Checking
- Feedback Collection
- Networking History Management
- User-Friendly Interface
- FastAPI REST API
- Streamlit Dashboard
- Gemini AI Integration

SYSTEM ARCHITECTURE

The Networking Assistant follows a three-tier architecture.

Presentation Layer

- Streamlit Web Application
- User Interface

Application Layer

- FastAPI Backend
- Event Analysis Module
- Topic Generation Module
- Fact Checking Module
- Feedback Module
- History Module

AI Layer

- Google Gemini API
- Prompt Engineering
- Response Generation

TECHNOLOGY STACK

Frontend

- Streamlit
- HTML
- CSS

Backend

- FastAPI
- Python

AI

- Google Gemini API

Database / Storage

- JSON File Storage

Development Tools

- VS Code
- Git
- GitHub
- Python Virtual Environment

WORKFLOW

1. User enters networking event details.
2. The request is sent to the FastAPI backend.
3. Gemini AI analyzes the event description.
4. AI generates networking topics and conversation suggestions.
5. Fact-checking validates user queries.
6. Feedback is collected from users.
7. Event history is stored.
8. Results are displayed on the Streamlit dashboard.

MODULES

Event Analysis Module

Analyzes the event description and extracts important themes for networking preparation.

Topic Generation Module

Generates intelligent discussion topics and conversation starters based on the event.

Fact Checking Module

Verifies factual information using Generative AI.

Feedback Module

Collects user feedback to improve the overall experience.

History Module

Stores networking history and allows users to review previous interactions.

FUTURE ENHANCEMENTS

- User Authentication
- Cloud Database Integration
- Resume Analysis
- LinkedIn Integration
- Voice-Based AI Assistant
- Multi-language Support
- Mobile Application
- Personalized Networking Recommendations
- Calendar Integration
- AI Email Draft Generation

CONCLUSION

Networking Assistant is an intelligent web application that simplifies networking preparation using Generative AI. The project helps users generate relevant conversation topics, verify information, organize networking history, and improve communication skills. By integrating FastAPI, Streamlit, and Google's Gemini AI, the application provides an efficient, scalable, and user-friendly solution for students and professionals participating in networking events.